

would conjure up a vision of a futuristic endeavour. A little disappointing when it's explained, right?

Still, the positronic brain quickly became the gold standard of artificial intelligence in science fiction. It would be made out of an alloy of platinum and iridium, while the shape could be left to the scientist. Asimov has suggested that it would need to look similar to the shape of a human brain, since the positronic brain would be based on the same neural network that the human brain runs on.

So be it R. Daneel Olivaw (who appears in several books), Andrew Martin of *The Positronic Man* (or as the film is called, *The Bicentennial Man*), or the VIKI computer of *I Robot*, each android in Asimov's fiction is powered by this positronic brain. However, Asimov was also quick to note that the hardware structure of the positronic brain was secondary to the software, namely the Three Laws of Robotics. Indeed, he suggested that to make a positronic brain that ignores the three laws would be almost impossible, requiring a complete rethinking of the positronic brain.

The Three Laws of Robotics

Perhaps no piece of science fiction has been as influential in actual artificial intelligence studies as the Three Laws of Robotics as proposed by Asimov. Indeed, scholarly professors argue about the Three Laws at seminars and John McCarthy himself once said that it was the clearest set of rules for future “neat AI” to be used in the service of man.

Asimov's Three Laws, first introduced in the 1942 short story *Runaround*, are as follows:



The Positronic Man?